

Class 17 Finish group material

Goals

1. Work with Cayley tables.
2. Compute order of elements.
3. Determine center and centralizers.

Definition:

G is an Abelian group means $ab = ba$ for all elements a, b in G .

Examples of Abelian groups

$$\mathbb{Z}_n, U(n), (\mathbb{Z}, +), (\mathbb{R}^*, \cdot)$$

Examples of nonabelian groups

$$S_n, D_n$$

More definitions:

The centralizer of an element g in group G is the set of all x in G such that $x \circ g = g \circ x$.

Elements that commute specifically with g .
Denoted $C(g)$

The center of a group is the set of all x in G such that $x \circ g = g \circ x$ for every g in G .
Elements that commute with everything.
 $Z(G)$

Use the Cayley table to answer the questions below.

	1	A	B	C	D	E	F	G
1	1	A	B	C	D	E	F	G
A	A	B	C	1	G	F	D	E
B	B	C	1	A	E	D	G	F
C	C	1	A	B	F	G	E	D
D	D	F	E	G	1	B	A	C
E	E	G	D	F	B	1	C	A
F	F	E	G	D	C	A	1	B
G	G	D	F	E	A	C	B	1

1. What is the identity element?

1

2. What is A^{-1} ?

C

3. What is the order of the group?

8

4. What is the order of the element C?

4

4. What is the center of this group?

$\{1, B\}$

5. what is the centralizer of D?

$\{1, B, D, E\}$

Compare the Cayley tables for $\mathbb{Z}_6$ and for $U(7)$.

$\mathbb{Z}_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$U(7)$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

Find the order of each element in $\mathbb{Z}_6$ and $U(7)$

$\mathbb{Z}_6$ and in $U(7)$

$\mathbb{Z}_6$	0	1	2	3	4	5
order	1	6	3	2	3	6

$U(7)$	1	2	3	4	5	6
order	1	3	6	3	6	2

$U(7)$	1	3	2	6	4	5
1	1	3	2	6	4	5
3	3	2	6	4	5	1
2	2	6	4	5	1	3
6	6	4	5	1	3	2
4	4	5	1	3	2	6
5	5	1	3	2	6	4

$$0 \rightarrow 1$$

$$1 \rightarrow 3$$

$$2 \rightarrow 2$$

$$3 \rightarrow 6$$

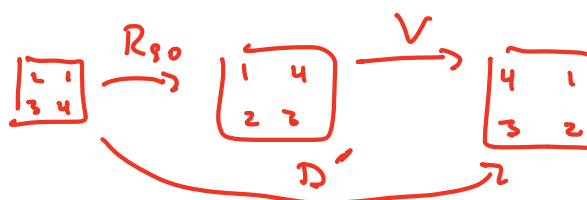
$$4 \rightarrow 4$$

$$5 \rightarrow 5$$

7. (6 pts) Shown below is a blank Cayley table for the group D_4 . Write the elements in cells labeled a and b as a product of elements in D_4 . Then draw diagrams to determine exactly which element should be in the cell. You must show your diagrams to receive full credit.

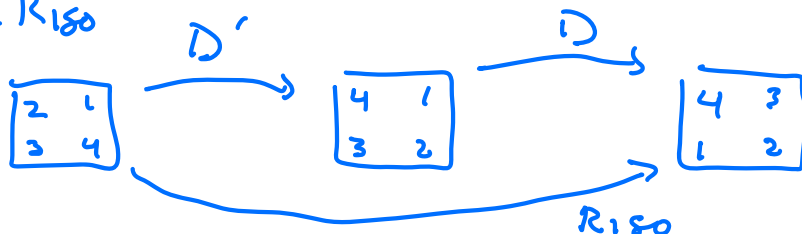
a = $V \circ R_{90} = D'$

Diagram for a:



b = $D \circ D' = R_{180}$

Diagram for b:



	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_0								
R_{90}								
R_{180}								
R_{270}								
H								
V		a						
D								b
D'								



H



V



D



D'

